Roll No. | | | | | | | |

End Semester Examination June 2025

Name of the Course: M. Tech **Semester: II**
Name of the Course: Earthquake resistant design of buildings **Course Code: MSE 201**
Time: 3 hours **Maximum Marks: 100**

Note:-
(i) **Answer any two parts from each question.**
(ii) **Each question carries 10 marks**

Q1	**(20marks)**	
(a)	Discuss the processing of strong ground motion data? Mention the role of response spectrum in designing earthquake resistant RC buildings?	
(b)	Which factors influence the intensity of earthquake at a particular place? Define the intensity of earthquake and explain how the intensity can be measured in Modified Mercalli Intensity Scale based on observations?	**CO1**
(c)	Explain elastic rebound theory and plate tectonic theory responsible for formation of earthquake with help of appropriate diagrams?	
Q2	**(20 marks)**	
(a)	Elaborate at least three seismic design philosophy used in ERD of buildings?	
(b)	Mention the general RC building construction practices used for design and construction of various structural members in a building?	**CO2**
(c)	Discuss various vertical irregularities in structural design of RC buildings and mention their effect on buildings during earthquake?	
Q3	**(20 marks)**	
(a)	Consider a 4 storey RC office building located in Roorkee. The soil conditions are medium stiff and the entire building is supported on a raft. The R.C. frames are infilled with brick-masonry (SMRF). The lumped weight due to dead loads is 15 kN/m^2 on floors and 12 kN/m^2 on the roof. The floors are to cater for a live load of 3 kN/m^2 on floors and 2.5 kN/m on the roof. Determine design seismic load on the structure as per new code. Plan area of the floors is 350 sq.m. and the height of each floor is 3.5m.	**CO3**
(b)	How can we obtain response spectrum from the responses of spring mass oscillators having different natural frequencies embedded in the ground having same damping?	
(c)	Explain all the possible failure mechanism of infilled wall frames with diagrams?	
Q4	**(20 marks)**	
(a)	Discuss the role of ductility in earthquake resistant design of buildings and how it can be improved in individual structural members as well as whole structure?	
(b)	What do you mean by special confining reinforcement and what are the requirements for such reinforcement mentioned in IS 13920: 2016?	**CO4**
(c)	What are the codal provisions for ductile detailing of longitudinal and web reinforcement in flexural members?	

Q5	(20 marks)	
(a)	Elaborate the purpose, principles and advantages of base isolation?	CO5
(b)	Mention various conventional and non- conventional strategies for retrofitting of structures?	
(c)	Discuss various sources of weakness in RC frame buildings?	